

Quiz #4A

Q1: $F(A, B, C, D) = \sum(4, 5, 8, 9, 14, 15)$

$d(A, B, C, D) = \sum(2, 3, 12, 13)$

		CD			
		00	01	11	10
AB	$\bar{C}\bar{D}$	$\bar{C}\bar{D}$	$\bar{C}D$	$C\bar{D}$	CD
	00 $\bar{A}\bar{B}$	0	0	X	X
	01 $\bar{A}B$	1	1	0	0
	11 $\bar{A}B$	X	X	1	1
	10 $A\bar{B}$	1	1	0	0

$F = (A+B)(A+\bar{C})(B+\bar{C})$

$F = B\bar{C} + A\bar{C} + AB$

Q2: $(\bar{A} + \bar{B} + C)(\bar{A} + \bar{B} + \bar{C})(\bar{B} + \bar{C} + D)(\bar{A}B\bar{C})(\bar{C}\bar{D}B)$

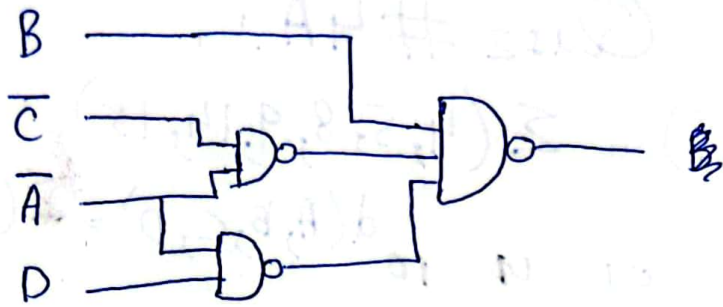
		CD			
		00	01	11	10
AB	$\bar{C}\bar{D}$	$\bar{C}\bar{D}$	$\bar{C}D$	$C\bar{D}$	CD
	(A+B) 00 $\bar{A}\bar{B}$	1	1	1	1
	(A+B) 01 $\bar{A}B$	1	1	1	0
	(A+B) 11 AB	0	0	0	0
	(A+B) 10 $A\bar{B}$	1	1	1	1

$$F = \bar{B} + \bar{A}\bar{C} + \bar{A}D$$

$$= \overline{(\overline{(\bar{B} + \bar{A}\bar{C} + \bar{A}D)})}$$

$$= \overline{B \cdot (\bar{A}\bar{C}) \cdot (\bar{A}D)}$$

ANALYZE



Q3:

$$(\bar{A}\bar{B} + A\bar{B})(CD + C\bar{D}) = \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D}$$

		CD			
		00	01	11	10
AB	$\bar{C}\bar{D}$				
	$\bar{C}D$				
	CD				
	$C\bar{D}$				
00	$\bar{A}\bar{B}$	0	1	1	1
01	$\bar{A}B$	4	5	7	6
11	AB	12	13	15	14
10	$A\bar{B}$	8	9	11	10

~~$\bar{B} + C$~~ $\bar{B}C$

Q4:

A	B	C	$\bar{A}+B$	$\bar{B}+\bar{C}$	$\bar{C}+A$	$(\bar{A}+B)(\bar{B}+\bar{C})(\bar{C}+A)$
0	0	0	1	1	1	1
0	0	1	1	1	0	0
0	1	0	1	1	1	1
0	1	1	1	0	0	0
1	0	0	0	1	1	0
1	0	1	0	1	1	0
1	1	0	1	1	1	1
1	1	1	1	0	1	0

Quiz #4B

Q1: $F = \Sigma(1, 2, 4, 6, 12)$ $d = \Sigma(0, 3, 5, 7, 11, 15)$

		CD			
AB		$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$(A+B)$	$\bar{A}\bar{B}$	X ₀	1 ₁	X ₃	1 ₂
$(A+B)$	$\bar{A}B$	1 ₄	X ₅	X ₇	1 ₆
$(\bar{A}+B)$	AB	1 ₁₂	0 ₁₃	X ₁₅	0 ₁₄
$(\bar{A}+B)$	$A\bar{B}$	0 ₈	0 ₉	X ₁₁	0 ₁₀

$$F = \bar{A} + B\bar{C}\bar{D}$$

$$F = (\bar{A} + \bar{C})(\bar{A} + \bar{D})(\bar{A} + B + \bar{C})$$

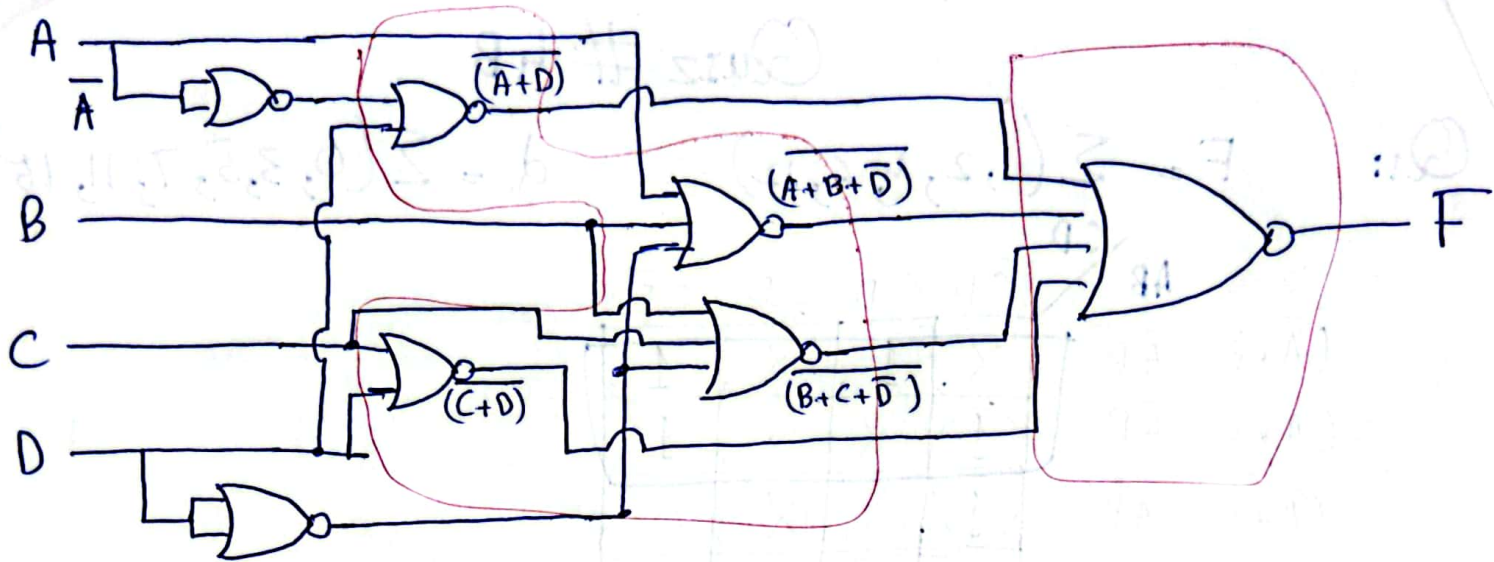
Q2: $\bar{A}C\bar{D} + \bar{A}BD + ABD + (\bar{A} + B + \bar{C} + \bar{D}) + (A + \bar{B} + \bar{D})$
 $\bar{A}C\bar{D} + \bar{A}BD + ABD + A\bar{B}CD + \bar{A}BD$

		CD			
AB		$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$(A+B)$	$\bar{A}\bar{B}$	0 ₀	0 ₁	0 ₃	1 ₂
$(A+B)$	$\bar{A}B$	0 ₄	1 ₅	1 ₇	1 ₆
$(\bar{A}+B)$	AB	0 ₁₂	1 ₁₃	1 ₁₅	0 ₁₄
$(\bar{A}+B)$	$A\bar{B}$	0 ₈	0 ₉	1 ₁₁	0 ₁₀

$$F = (C + D)(\bar{A} + D)(A + B + \bar{D})(B + C + \bar{D})$$

$$= \overline{((C + D)(\bar{A} + D)(A + B + \bar{D})(B + C + \bar{D}))}$$

$$= \overline{((\overline{C + D}) + (\overline{\bar{A} + D}) + (\overline{A + B + \bar{D}}) + (\overline{B + C + \bar{D}}))}$$



Q3: $(C+CD)(C+\bar{C}D)(C+E)$

$$[C(1+D)] [\cancel{C}C + \bar{C}D] [C+E]$$

$$(C) (C+D)(C+E)$$

↓ Absorption law

$$= (CC + CD)(C+E) \quad \because \text{distributive law}$$

$$= (C+CD)(C+E)$$

$$= (C)(C+E) \quad \because \text{Absorption law}$$

$$= CC + CE \quad \because \text{Distributive law}$$

$$= C + CE = C \quad \because \text{Absorption law}$$

Q4: $(\bar{A}+B)(\bar{A}+\bar{B}+\bar{C})(B+\bar{C}+D)(A+\bar{B}+C+\bar{D})$

CD

AB	$(C+D)$	$(C+\bar{D})$	$(\bar{C}+\bar{D})$	$(\bar{C}+D)$
$(\bar{A}+B)$	0	1	3	2
$\bar{A}+\bar{B}$	4	5	7	6
$(\bar{A}+\bar{B})$	12	13	15	14
$(\bar{A}+B)$	8	9	11	10

$$F = (\bar{A}+B)(\bar{A}+\bar{C})(B+\bar{C}+D)(\bar{A}+\bar{B}+C+\bar{D})$$